

Jianning Li, Ph.D.

Lecturer & Senior Researcher in Medical AI

Free University of Berlin (FU Berlin) • Charité – Universitätsmedizin Berlin • Zuse-Institut Berlin (ZIB)

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RESEARCH OVERVIEW & EXPERTISE

Computer scientist specializing in medical image analysis, deep generative modeling, and 3D computer vision. Core research focuses on interpretable and disentangled representation learning, automated cranial & anatomical shape reconstruction, dental AI morphology, and translating deep learning frameworks into scalable, reproducible clinical decision support systems.

ACADEMIC & RESEARCH POSITIONS

Lecturer & Academic Instructor

Ongoing

[Free University of Berlin \(FU Berlin\)](#) & [Charité – Universitätsmedizin Berlin](#) • Berlin, Germany

Lecturing and co-instructing courses on Machine Learning, Data Science for Life Science, and specialized seminars on Large Language Models, Agentic AI, and Dental AI.

Researcher

Ongoing

[Zuse-Institut Berlin \(ZIB\)](#) • Berlin, Germany

Conducting research in medical computer vision, 3D shape reconstruction, uncertainty estimation, and open-source medical software pipelines.

EDUCATION

Ph.D. in Computer Science

With Distinction

Focus: Medical Image Analysis, 3D Deep Generative Models & Cranial Implant Design

M.S. in Computer Science

With Distinction

Focus: Computer Vision, Machine Learning, Signal & Image Processing

B.S. in Biomedical Engineering

With Distinction

Focus: Medical Physics, Biomechanics, Clinical Engineering

SELECTED PUBLICATIONS

[Li, J.](#), Bitter, K., AL, M.A., Nguyen, A.D., Shemesh, H., Zaslansky, P., Zachow, S. "Comparative Analysis of 3D Root Canal Structures using Micro-CT." *Oral Health Research Congress (CED-IADR)*, 2026.

[Li, J.](#), Sengupta, A., Zachow, S. "Uncertainty estimation and probabilistic skull shape reconstruction using Bayesian neural networks." *Scientific Reports*, 16, 16383, 2026.

Sengupta, A., Kuş, D., [Li, J.](#), Zachow, S. "Globally Optimal Pose from Orthographic Silhouettes." *IEEE/CVF CVPR 2026*.

[Li, J.](#), Bitter, K., et al. "Computational Insights into Root Canal Treatment: A Survey of Selected Methods in Imaging, Segmentation, Morphological Analysis, and Clinical Management." *Dentistry Journal*, 13(12), 2025. [Featured Cover Paper](#)

[Li, J.](#), Zhou, Z., Yang, J., Pepe, A., Gsaxner, C., Luijten, G., et al. "MedShapeNet – a large-scale dataset of 3D medical shapes for computer vision." *Biomedical Engineering / Biomedizinische Technik*, 70(1), 2025.

Eisenmann, M., Reinke, A., ..., [Li, J.](#), ..., et al. "Why is the winner the best?" *IEEE/CVF CVPR 2023*, pp. 19955–19966, 2023.

Ferreira, A., Solak, N., [Li, J.](#), Dammann, P., Kleesiek, J., Alves, V., Egger, J. "How we won BraTS 2023 Adult Glioma challenge? Enhanced Synthetic Data Augmentation and Model Ensemble for brain tumour segmentation." *BraTS 2023*. [First Place Winner](#)

[Li, J.](#), Ellis, D.G., Kodym, O., Rauschenbach, L., Rieß, et al. "Towards clinical applicability and computational efficiency in automatic cranial implant design: An overview of the AutoImplant 2021 cranial implant design challenge." *Medical Image Analysis (MedIA)*, 88, 102865, 2023.

Gsaxner, C., [Li, J.](#), Pepe, A., et al. "The HoloLens in medicine: A systematic review and taxonomy." *Medical Image Analysis (MedIA)*, 85, 102757, 2023.

[Li, J.](#), Pepe, A., Luijten, G., Schwarz-Gsaxner, C., et al. "Anatomy Completor: A Multi-class Completion Framework for 3D Anatomy Reconstruction." *ShapeMI @ MICCAI 2023*, pp. 1–14, 2023.

[Li, J.](#), Gsaxner, C., Pepe, A., et al. "Sparse convolutional neural network for high-resolution skull shape completion and shape super-resolution." *Scientific Reports*, 2023.

[Li, J.](#), Fragemann, J., et al. "Training β -VAE by Aggregating a Learned Gaussian Posterior with a Decoupled Decoder." *MAD @ MICCAI 2022*, Springer.

Li, J., Pimentel, P., Szengel, A., Ehlke, M., Lamecker, H., et al. "AutoImplant 2020 — First MICCAI Challenge on Automatic Cranial Implant Design." *IEEE Transactions on Medical Imaging (TMI)*, 2021.

Li, J., von Campe, G., Pepe, A., Gsaxner, C., et al. "Automatic Skull Defect Restoration and Cranial Implant Generation for Cranioplasty." *Medical Image Analysis (MedIA)*, 2021.

Pepe, A.*, **Li, J.*** (*Equal contribution), et al. "Detection, segmentation, simulation and visualization of aortic dissections: A review." *Medical Image Analysis (MedIA)*, 65, 101773, 2020.

BOOKS & PROCEEDINGS (EDITOR)

- **Shape in Medical Imaging** • Proceedings of MICCAI Workshop (ShapeMI 2023), Springer LNCS, 2023.
- **Medical Applications with Disentanglements** • Proceedings of MICCAI Workshop (MAD 2022), Springer LNCS, 2022.
- **Towards the Automatization of Cranial Implant Design in Cranioplasty II** • AutoImplant II @ MICCAI 2021, Springer, 2021.
- **Towards the Automatization of Cranial Implant Design in Cranioplasty I** • AutoImplant I @ MICCAI 2020, Springer, 2020.

KEY TALKS & PRESENTATIONS

- **Invited Talk:** "MedShapeNet: A Large Scale Dataset of 3D Medical Shapes for Computer Vision" • CCVL Lab, Johns Hopkins University (JHU), 2023.
- **Invited Talk:** "MedShapeNet in Coronary Artery & Vascular Segmentation" • IEEE EMBC Workshop, 2023.
- **Oral Presentation:** "Anatomy Completor: A Multi-class Completion Framework for 3D Anatomy Reconstruction" • ShapeMI @ MICCAI 2023.
- **Invited Talk:** "Medical Shape Completion" • Institute for AI in Medicine (IKIM), 2022.
- **Lecture:** "Medical 3D Printing & Clinical AI" • Wahlfach Klinik: Digitale Medizin und KI, 2022.

ACADEMIC SERVICE & LEADERSHIP

Challenge Lead & Organizer:

- AutoImplant I (MICCAI 2020) & AutoImplant II (MICCAI 2021)

Workshop Co-Organizer:

- MedShapeNet Tutorial (MICCAI 2024)
- ShapeMI 2023 (MICCAI 2023)
- MAD 2022 (MICCAI 2022)

Peer Reviewer for Top Journals & Conferences:

- MICCAI • IEEE Transactions on Medical Imaging (TMI)
- Medical Image Analysis (MedIA) • IEEE ISBI
- Computers in Biology and Medicine

TEACHING & STUDENT MENTORSHIP

- **Lecturer:** *Digitale Medizin und Künstliche Intelligenz* (Essen)
- **Co-instructor:** *Data Science for Life Science & Machine Learning* (FU Berlin)
- **Specialized Seminars & Workshops:** *LLMs, Agentic AI, and Dental AI Assistants* (Charité)
- **Supervision:** Active mentor for Bachelor's & Master's thesis projects in 3D medical image segmentation, shape reconstruction, and interpretable AI.

FEATURED PROJECTS & OPEN SOURCE SOFTWARE

- **MONAI Skull Reconstruction:** Open-source pipeline built on MONAI for reproducible cranial implant design.
- **MedShapeNet Platform:** Large-scale 3D medical shape dataset and interactive cloud platform for 3D computer vision research.
- **Private Local Dental AI Assistant:** Local desktop AI application designed for automated dental workflow assistance.